
AS
FURTHER MATHEMATICS
7366/1

Paper 1

Mark scheme

June 2021

Version: 1.0 Final Mark Scheme



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Mark scheme instructions to examiners

General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- marking instructions that indicate when marks should be awarded or withheld including the principle on which each mark is awarded. Information is included to help the examiner make his or her judgement and to delineate what is creditworthy from that not worthy of credit
- a typical solution. This response is one we expect to see frequently. However credit must be given on the basis of the marking instructions.

If a student uses a method which is not explicitly covered by the marking instructions the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

Key to mark types

M	mark is for method
R	mark is for reasoning
A	mark is dependent on M marks and is for accuracy
B	mark is independent of M marks and is for method and accuracy
E	mark is for explanation
F	follow through from previous incorrect result

Key to mark scheme abbreviations

CAO	correct answer only
CSO	correct solution only
ft	follow through from previous incorrect result
'their'	indicates that credit can be given from previous incorrect result
AWFW	anything which falls within
AWRT	anything which rounds to
ACF	any correct form
AG	answer given
SC	special case
OE	or equivalent
NMS	no method shown
PI	possibly implied
sf	significant figure(s)
dp	decimal place(s)

Examiners should consistently apply the following general marking principles:

No Method Shown

Where the question specifically requires a particular method to be used, we must usually see evidence of use of this method for any marks to be awarded.

Where the answer can be reasonably obtained without showing working and it is very unlikely that the correct answer can be obtained by using an incorrect method, we must award **full marks**. However, the obvious penalty to candidates showing no working is that incorrect answers, however close, earn **no marks**.

Where a question asks the candidate to state or write down a result, no method need be shown for full marks.

Where the permitted calculator has functions which reasonably allow the solution of the question directly, the correct answer without working earns **full marks**, unless it is given to less than the degree of accuracy accepted in the mark scheme, when it gains **no marks**.

Otherwise we require evidence of a correct method for any marks to be awarded.

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Work erased or crossed out

Erased or crossed out work that is still legible and has not been replaced should be marked. Erased or crossed out work that has been replaced can be ignored.

Choice

When a choice of answers and/or methods is given and the student has not clearly indicated which answer they want to be marked, mark positively, awarding marks for all of the student's best attempts. Withhold marks for final accuracy and conclusions if there are conflicting complete answers or when an incorrect solution (or part thereof) is referred to in the final answer.

AS/A-level Maths/Further Maths assessment objectives

AO		Description
AO1	AO1.1a	Select routine procedures
	AO1.1b	Correctly carry out routine procedures
	AO1.2	Accurately recall facts, terminology and definitions
AO2	AO2.1	Construct rigorous mathematical arguments (including proofs)
	AO2.2a	Make deductions
	AO2.2b	Make inferences
	AO2.3	Assess the validity of mathematical arguments
	AO2.4	Explain their reasoning
	AO2.5	Use mathematical language and notation correctly
AO3	AO3.1a	Translate problems in mathematical contexts into mathematical processes
	AO3.1b	Translate problems in non-mathematical contexts into mathematical processes
	AO3.2a	Interpret solutions to problems in their original context
	AO3.2b	Where appropriate, evaluate the accuracy and limitations of solutions to problems
	AO3.3	Translate situations in context into mathematical models
	AO3.4	Use mathematical models
	AO3.5a	Evaluate the outcomes of modelling in context
	AO3.5b	Recognise the limitations of models
	AO3.5c	Where appropriate, explain how to refine models

Q	Marking instructions	AO	Marks	Typical solution
1	Ticks correct box	1.1b	B1	$\cos(2) + i \sin(2)$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
2	Circles correct answer	1.1b	B1	17
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
3	Ticks correct box	1.1b	B1	$a = -\frac{1}{2} \quad b = -\frac{\sqrt{3}}{2}$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
4	Circles correct answer	1.1b	B1	$\begin{bmatrix} 4 & 6 \\ 2 & 5 \end{bmatrix}$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
5	Calculates the scalar product of the two vectors. Accept one miscopy of an element, or one incorrect product if the elements are not shown.	1.1a	M1	$\begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ 4 \\ 1 \end{bmatrix} = 1 \times 7 + (-3) \times 4 + 5 \times 1$ $= 7 - 12 + 5 = 0$ $\therefore$ the two vectors are perpendicular
	Completes a fully correct proof to reach the required result. Must include at least one calculation line and $=0$ linked with perpendicular conclusion. Ignore incorrect vector magnitudes.	2.1	R1	
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
6	Recalls exponential definitions of hyperbolic functions and substitutes $\frac{e^x + e^{-x}}{2}$ and $\frac{e^x - e^{-x}}{2}$ Condone cosh x replaced with $\frac{e^x + e^{-x}}{2}$ and sinh x replaced with $\frac{e^x - e^{-x}}{2}$	1.2	M1	$\cosh^2 x - \sinh^2 x =$ $\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$ $= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4}$ $= \frac{4}{4}$ $= 1$
	Completes a fully correct proof to reach the required result.	2.1	R1	
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
7	Writes the given expression in the form $m + \ln(1 + f(x))$ where m is a non-zero constant (allow unsimplified). Or obtains the first three derivatives of $\ln(e + 2ex)$ in the form $p(q + rx)^n$	3.1a	M1	$\begin{aligned}\ln(e + 2ex) &= \ln(e(1 + 2x)) \\ &= \ln e + \ln(1 + 2x) \\ &= 1 + 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \dots \\ &= 1 + 2x - 2x^2 + \frac{8}{3}x^3 - \dots\end{aligned}$
	From an expression of the form $\ln(1 + f(x))$, substitutes $f(x)$ into the Maclaurin series. Condone sign errors and missing brackets. Or substitutes $x = 0$ into their $f(0)$, $f'(0)$, $f''(0)$ and $f'''(0)$ of the form $p(q + rx)^n$	1.1a	M1	
	Completes a fully correct argument to reach the required result and correctly calculates a . Do not condone missing brackets in their working.	2.1	R1	
Total			3	

Q	Marking instructions	AO	Marks	Typical solution
8(a)	Gives a correct explanation. Condone an explanation which suggests that conjugate pairs <u>only</u> occur if the coefficients are real.	2.3	B1	Can only assume that complex roots of a polynomial equation are in conjugate pairs if the coefficients are all real.
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
8(b)	Writes an equation for the 3 rd root using the sum of roots. Condone $-2i$ or 2 or -2 for the sum of roots. Or writes an equation in p and q (may be unsimplified) from substitution of one known root into the equation.	3.1a	M1	$(1 + i) + (-1) + \alpha = 2i$ $\alpha = i$ $p = (1 + i)(-1) + (-1)i + i(1 + i)$ $p = -1 - i - i + i + i^2$ $p = -2 - i$ $q = -(1 + i)(-1)i$ $q = i + i^2$ $q = i - 1$
	Obtains correct 3 rd root. PI by a factor of $(z - i)$ Or writes two correct simultaneous equations in p and q from substitution of both known roots into the equation (any i^2 must be replaced with -1).	1.1b	A1	
	Writes an expression for p using the sum of pairwise products of roots with their 3 rd root and the two given roots. Or eliminates q from their simultaneous equations. Or multiplies $(z - (1 + i))(z + 1)(z - \alpha)$ to obtain a cubic expression where α is their 3 rd root. May be unsimplified. Allow sign errors.	3.1a	M1	
	Writes an expression for q using the product of roots with their 3 rd root and the two given roots. Or eliminates p from their simultaneous equations. Or compares the coefficients of the given polynomial and their cubic expansion. Allow sign errors.	3.1a	M1	
	Obtains $p = -2 - i$ and $q = i - 1$	1.1b	A1	
Total			5	

Question total			6	
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Q	Marking instructions	AO	Marks	Typical solution
9(a)	Writes the sum in the form $\alpha \sum r^2 + \beta \sum r$ where α and β are numbers. PI by use of correct standard formulae.	1.1a	M1	$\sum_{r=1}^n r(r+3) = \sum_{r=1}^n r^2 + 3 \sum_{r=1}^n r$ $= \frac{1}{6}n(n+1)(2n+1) + 3 \times \frac{1}{2}n(n+1)$ $= \frac{1}{6}n(n+1)((2n+1) + 9)$ $= \frac{1}{6}n(n+1)(2n+10)$ $= \frac{1}{3}n(n+1)(n+5)$
	Recalls and uses standard formulae to obtain a correct expression for the sum in terms of n May be unsimplified.	1.2	A1	
	Identifies n and $(n+1)$ as common factors. Must be at least one correct term inside the remaining bracket.	1.1a	M1	
	Completes a fully correct proof to reach the required result, including a clear statement that the original sum can be written in the form $\alpha \sum r^2 + \beta \sum r$ Must have $a = \frac{1}{3}$ and $b = 5$	2.1	R1	
Total			4	

Q	Marking instructions	AO	Marks	Typical solution
9(b)	Substitutes $5n$ into their expression of the form $an(n+1)(n+b)$ and subtracts their part (a) .	1.1a	M1	$\sum_{r=n+1}^{5n} r(r+3)$ $= \sum_{r=1}^{5n} r(r+3) - \sum_{r=1}^n r(r+3)$ $= \frac{1}{3}5n(5n+1)(5n+5) - \frac{1}{3}n(n+1)(n+5)$ $= \frac{1}{3}n(n+1)(25(5n+1) - (n+5))$ $= \frac{1}{3}n(n+1)(124n+20)$ $= \frac{4}{3}n(n+1)(31n+5)$
	Obtains a correct expression. May be unsimplified.	1.1b	A1	
	Obtains a correct fully factorised expression.	1.1b	A1	
Total			3	

Question total			7	
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Q	Marking instructions	AO	Marks	Typical solution
10(a)	Writes a correct unsimplified expression for $\det \mathbf{A}$	1.1a	M1	$3 \times 2 - i(i - 1) = 6 - i^2 + i$ $= 7 + i$
	Completes a fully correct proof to reach the required result. Must have $a = 7$	2.1	R1	
	Total		2	

Q	Marking instructions	AO	Marks	Typical solution
10(b)	Recognises that B is equal to a multiple of A ⁻¹ Or multiplies A and B to find at least one correct unsimplified element of AB	3.1a	M1	$\mathbf{B} = p\mathbf{A}^{-1}$ $\begin{bmatrix} 14 - 2i & b \\ c & d \end{bmatrix} = \frac{p}{7 + i} \begin{bmatrix} 2 & 1 - i \\ -i & 3 \end{bmatrix}$ $14 - 2i = \frac{2p}{7 + i}$ $(7 - i)(7 + i) = p$ $p = 50$ $b = \frac{50(1 - i)}{7 + i} = 6 - 8i$ $c = \frac{-50i}{7 + i} = -1 - 7i$ $d = \frac{150}{7 + i} = 21 - 3i$
	Sets up at least one correct non-matrix equation in one or two unknowns by equating a pair of corresponding elements.	1.1a	M1	
	Sets up another correct non-matrix equation in one unknown only.	1.1a	M1	
	Obtains at least one correct value of p, b, c or d . Could be seen as an element of a matrix, e.g. $\frac{1}{50} \begin{bmatrix} 14 - 2i & * \\ * & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & 6 - 8i \\ * & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & * \\ -1 - 7i & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & * \\ * & 21 - 3i \end{bmatrix}$	1.1b	A1	
	Obtains at least two correct values of p, b, c or d . Could be seen as an element of a matrix, e.g. $\frac{1}{50} \begin{bmatrix} 14 - 2i & * \\ * & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & 6 - 8i \\ * & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & * \\ -1 - 7i & * \end{bmatrix}$ or $k \begin{bmatrix} 14 - 2i & * \\ * & 21 - 3i \end{bmatrix}$	1.1b	A1	
	Obtains all four correct values of p, b, c and d . Accept $p = 50$ and $\mathbf{B} = \begin{bmatrix} 14 - 2i & 6 - 8i \\ -1 - 7i & 21 - 3i \end{bmatrix}$	1.1b	A1	
Total			6	

Question total	8
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Q	Marking instructions	AO	Marks	Typical solution
11(a)	Obtains the correct result including at least one intermediate step. All lines must be correct. Condone original expression omitted.	2.1	B1	$\frac{1}{(r-1)!} - \frac{1}{r!} = \frac{r}{r!} - \frac{1}{r!} = \frac{r-1}{r!}$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
11(b)	Writes the first two pairs (or last two pairs) of corresponding terms of $\frac{1}{(r-1)!}$ and $\frac{1}{r!}$	1.1a	M1	$\sum_{r=1}^n \frac{r-1}{r!} = \sum_{r=1}^n \left(\frac{1}{(r-1)!} - \frac{1}{r!} \right)$ $= \frac{1}{0!} - \frac{1}{1!}$ $+ \frac{1}{1!} - \frac{1}{2!}$ $+ \dots$ $+ \frac{1}{(n-2)!} - \frac{1}{(n-1)!}$ $+ \frac{1}{(n-1)!} - \frac{1}{n!}$ $= \frac{1}{0!} - \frac{1}{n!}$ $= 1 - \frac{1}{n!}$
	Writes at least the first pair and the last pair of corresponding terms of $\frac{1}{(r-1)!}$ and $\frac{1}{r!}$ and shows the pattern of cancelling.	1.1a	M1	
	Completes a fully correct proof to reach the required result with $a = 1$ and $b = -1$ Must include the 1 st and n^{th} terms and at least one pair of cancelling terms when completing the method of differences process.	2.1	R1	
Total			3	

Question total			4	
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Q	Marking instructions	AO	Marks	Typical solution
12(a)	Obtains the other two roots.	1.1b	B1	$x^3 - 2x^2 - x + 2 = (x - 2)(x^2 - 1)$ $= (x - 2)(x - 1)(x + 1)$ $x = 1 \text{ and } x = -1$
	Total		1	

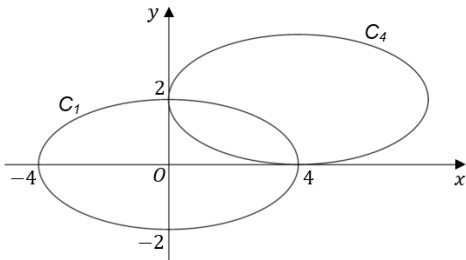
Q	Marking instructions	AO	Marks	Typical solution
12(b)	Obtains at least one correct value for $\cosh \theta$ Or uses their answer to part (a) to write down a value for $\cosh \theta$	1.1a	M1	$\cosh \theta = 2 ,$ $\cosh \theta = 1 ,$ $\cosh \theta = -1$ but $\cosh \theta \geq 1 \quad \therefore \cosh \theta \neq -1$ $\theta = \pm \ln(2 + \sqrt{3}) , \theta = 0$
	Rejects $\cosh \theta = -1$ Follow through their part (a) < 1 PI by only considering values for which $\cosh \theta \geq 1$	1.1b	B1ft	
	Correctly finds at least one non-zero root of the equation. Follow through any of their answers to part (a) if greater than 1.	1.1b	A1ft	
	Obtains the three correct values of θ with no incorrect answers. Accept $\theta = \pm \cosh^{-1}(2)$	1.1b	A1	
	Total		4	

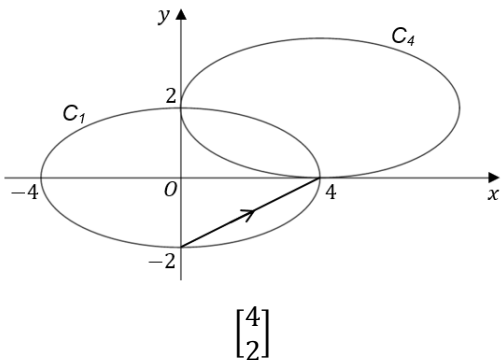
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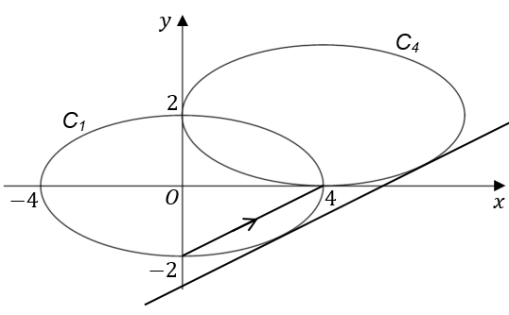
Q	Marking instructions	AO	Marks	Typical solution
13	Demonstrates the rule is correct for $n = 1$	2.2a	B1	Try $n = 1$: $\text{LHS} = 2^{-1} = \frac{1}{2}$ and $\text{RHS} = 1 - 2^{-1} = \frac{1}{2}$ $\text{LHS} = \text{RHS} \therefore \text{the rule is true for } n = 1$
	Assumes the rule is true for $n = k$ and adds $2^{-(k+1)}$ to $1 - 2^{-k}$ Condone incorrect or missing brackets for this mark only.	2.4	M1	
	Correctly obtains $1 - 2^{-(k+1)}$ from $1 - 2^{-k} + 2^{-(k+1)}$	2.2a	A1	Assume the rule is true for $n = k$: $\sum_{r=1}^k 2^{-r} = 1 - 2^{-k}$
	Concludes a reasoned argument by stating that The rule is true for $n = 1$; that if the rule is true for $n = k$ then it is also true for $n = k + 1$ and hence, by induction, the rule is true for all integers $n \geq 1$.	2.1	R1	$\sum_{r=1}^k 2^{-r} + 2^{-(k+1)} = 1 - 2^{-k} + 2^{-(k+1)}$ $\sum_{r=1}^{k+1} 2^{-r} = 1 + 2^{-(k+1)}(-2^1 + 1)$ $\sum_{r=1}^{k+1} 2^{-r} = 1 - 2^{-(k+1)}$ $\therefore \text{the rule is also true for } n = k + 1$ As the rule is true for $n = 1$, and if true for $n = k$ then it is also true for $n = k + 1$, then by induction the rule is true for all integers $n \geq 1$
Total			4	

Q	Marking instructions	AO	Marks	Typical solution
14(a)	Writes down a correct equation.	1.1b	B1	$\frac{y^2}{16} + \frac{x^2}{4} = 1$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
14(b)	Indicates a stretch. Condones a stretch in any direction.	3.1a	M1	$y \rightarrow \frac{y}{2}$
	Identifies the correct transformation.	1.1b	A1	stretch, parallel to the y-axis, scale factor 2
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
14(c)(i)	Draws one loop centred on the origin and a second loop, approximately the same shape as the first, in the 1 st quadrant with the positive x and y axes as tangents. Or draws one correct graph with one correct x -intercept and one correct y -intercept.	1.1a	M1	
	Draws two correct graphs with all four intercepts correctly indicated.	1.1b	A1	
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
14(c)(ii)	States a translation vector which contains either 2 or -2 and 4 or -4 Follow through their intercepts.	3.1a	M1	
	Obtains the correct translation vector.	1.1b	A1	
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
14(c)(iii)	Calculates $\frac{b}{a}$ or $\frac{a}{b}$ for their translation vector $\begin{bmatrix} a \\ b \end{bmatrix}$	3.1a	M1	 $m = \frac{2}{4} = \frac{1}{2}$
	Obtains the correct gradient. Follow through their part (cii)	2.2a	A1F	
	Total		2	
	Question total		9	

Q	Marking instructions	AO	Marks	Typical solution
15(a)(i)	Forms one equation by substituting two of $x = 2 + 5\lambda$, $y = -1 + 3\lambda$, $z = 4 - 2\lambda$ into the equation of the second line. Or introduces a second parameter (eg μ) to form at least two equations in λ and μ , e.g. $2 + 5\lambda = 4\mu + 5$, $-1 + 3\lambda = 2\mu$, $4 - 2\lambda = 4 - \mu$	1.1a	M1	$\frac{2 + 5\lambda - 5}{4} = \frac{-1 + 3\lambda}{2}$ $2(5\lambda - 3) = 4(3\lambda - 1)$ $10\lambda - 12 = -4 + 6$ $-2\lambda = 2$ $\lambda = -1$ $\frac{2 + 5\lambda - 5}{4} = 4 - (4 - 2\lambda)$ $5\lambda - 3 = 4 \times 2\lambda$ $-3 = 8\lambda - 5\lambda$ $-3 = 3\lambda$ $\lambda = -1$ same value of $\lambda \therefore$ the lines intersect
	Solves their equation(s) to obtain the correct value of λ (or μ).	1.1b	A1	
	Completes a correct argument to conclude that the lines intersect. eg shows that $\lambda = -1$ satisfies another equation formed by substituting a different combination of x , y and z Or shows that $\lambda = -1$ and $\mu = -2$ satisfy a third equation. Or shows that $(-3, -4, 6)$ lies on both lines.	2.1	R1	
Total			3	

Q	Marking instructions	AO	Marks	Typical solution
15(a)(ii)	Obtains $\begin{bmatrix} -3 \\ -4 \\ 6 \end{bmatrix}$ Accept $-3\mathbf{i} - 4\mathbf{j} + 6\mathbf{k}$ Condone $(-3, -4, 6)$	1.1b	B1	$\mathbf{r} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} - 1 \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$ $= \begin{bmatrix} -3 \\ -4 \\ 6 \end{bmatrix}$
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
15(b)	States a correct explanation.	3.2a	B1	The submarines might not be at the intersection point at the same time.
Total			1	

Q	Marking instructions	AO	Marks	Typical solution
15(c)	Calculates the scalar product of the two direction vectors. Allow one error in each direction vector.	1.1a	M1	$\begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix}$ $= 5 \times 4 + 3 \times 2 + (-2) \times (-1) = 28$ $\sqrt{5^2 + 3^2 + (-2)^2} \times \sqrt{4^2 + 2^2 + (-1)^2}$ $= \sqrt{38} \times 21 = \sqrt{798}$ $\cos \theta = \frac{28}{\sqrt{798}}$ $\theta = 7.6^\circ \text{ (1dp)}$
	Calculates the product of the direction vector magnitudes.	1.1a	M1	
	Obtains correct angle. Condone 0.13 (radians). Accept 7.61° or better.	1.1b	A1	
Total			3	

	Question total		8	
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Q	Marking instructions	AO	Marks	Typical solution
16(a)	Obtains $a = 3$ Accept $y = \frac{3x}{x+b}$ with any non-zero value of b	2.2a	B1	$y = \frac{3x}{x+2}$ $a = 3 \quad \text{and} \quad b = 2$
	Obtains $b = 2$ Accept $y = \frac{ax}{x+2}$ with any non-zero value of a	2.2a	B1	
Total			2	

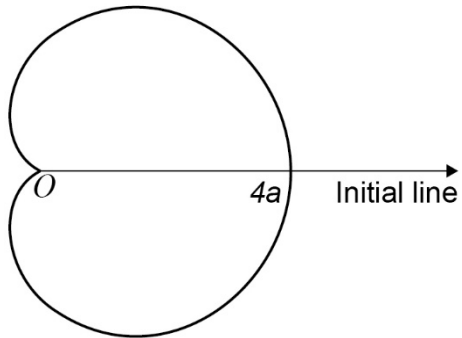
Q	Marking instructions	AO	Marks	Typical solution
16(b)	Explains that $y = \frac{3x}{2}$ is a tangent to the curve. Condone 'tangent' only.	2.4	B1	$y = \frac{3x}{2}$ is a tangent to the hyperbola. Any tangent to a hyperbola only intersects once. So there is exactly one root of the equation.
	Explains that a tangent to a hyperbola meets the curve exactly once. Accept 'conic' instead of 'hyperbola'. Condone no conclusion given.	2.4	B1	
Total			2	

Q	Marking instructions	AO	Marks	Typical solution
16(c)	Forms a simplified quadratic equation from $ax - (1 - x)(x + b) = 0$ Accept any inequality sign instead of = PI by $x^2 + 4x - 2$ or $-x^2 - 4x + 2$ PI by $-2 + \sqrt{6}$ or $-2 - \sqrt{6}$ Or forms a simplified cubic equation from $ax(x + 2) - (1 - x)(x + b)(x + 2) = 0$	1.1a	M1	$\frac{3x}{x + 2} = 1 - x$ $3x = (1 - x)(x + 2)$ $3x = -x^2 - x + 2$ $x^2 + 4x - 2 = 0$ $x = \frac{-4 \pm \sqrt{16 + 4 \times 2}}{2}$ $x = -2 \pm \sqrt{6}$ $x \leq -2 - \sqrt{6},$ $-2 < x \leq -2 + \sqrt{6}$
	Identifies $-2 + \sqrt{6}$ or $-2 - \sqrt{6}$ as a critical value. Follow through their a and b	1.1b	A1F	
	Identifies -2 as a critical value.	1.1b	B1	
	Obtains the correct set of values of x	3.2a	A1	
Total			4	
Question total			8	

Q	Marking instructions	AO	Marks	Typical solution
17(a)	States correct polar coordinates. Condone $(0, 2a)$ after seeing $r = 2a$ and $\theta = 0$	1.1b	B1	$r = 2a(1 + \sin 0) = 2a$ $M = (2a, 0)$
	Total		1	

Q	Marking instructions	AO	Marks	Typical solution
17(b)	Substitutes $\sin \theta = 1$ PI by $r = 4a$ or $\theta = \frac{\pi}{2}$	3.1a	M1	$r = 2a(1 + 1) = 4a$ $N = \left(4a, \frac{\pi}{2}\right)$
	Obtains the correct polar coordinates. Condone $\left(\frac{\pi}{2}, 4a\right)$ after $r = 4a$ and $\theta = \frac{\pi}{2}$ Condone $\left(\frac{\pi}{2}, 4a\right)$ if $(0, 2a)$ has been penalised in (a).	1.1b	A1	
	Total		2	

Q	Marking instructions	AO	Marks	Typical solution
17(c)	Forms an equation by equating the two polar equations.	3.1a	M1	$3a = 2a(1 + \sin \theta)$ $\frac{3}{2} = 1 + \sin \theta$
	Obtains at least one correct θ value for P or Q Implied by a correct area.	1.1b	A1	$\sin \theta = \frac{1}{2}$ $\theta = \frac{\pi}{6}, \quad \theta = \frac{5\pi}{6}$
	Calculates the x -coordinate of P or Q , ie $x = 3a \cos \theta$ Implied by $6a \cos \theta$ for the length PQ . Or calculates the y -coordinate of P or Q , ie $y = 3a \sin \theta$	1.1a	M1	$x = 3a \cos \frac{\pi}{6}$ $x = \frac{3\sqrt{3}a}{2}$
	Calculates the required area (or half of the required area). ie $\frac{1}{2} \times (y_N - y_P) \times 6a \cos \theta$	2.2a	M1	$y = 3a \sin \frac{\pi}{6}$ $y = \frac{3a}{2}$
	Obtains the correct area.	3.2a	A1	$\text{area} = \frac{1}{2} \times \frac{6\sqrt{3}a}{2} \times \left(4a - \frac{3a}{2}\right)$ $= \frac{3\sqrt{3}a}{2} \times \frac{5a}{2}$ $= \frac{15}{4}\sqrt{3}a^2$
Total			5	

Q	Marking instructions	AO	Marks	Typical solution
17(d)	Draws correct shape in each of the four quadrants with a cusp at the pole.	1.1b	B1	
	Marks the point $(4a, 0)$ on the diagram. Or marks $4a$ as the intercept on the initial line. Condone omission of $(0,0)$.	1.1b	B1	
	Total		2	

	Question total		10	
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	Paper total		80	
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